

Alex Hunter

Robotics & Embedded Systems Engineer

Glendenning, NSW; +61 424 980 107; alex.hunter4004@gmail.com
[linkedin.com/in/alexhunter4/](https://www.linkedin.com/in/alexhunter4/); github.com/alex-hunter-dev; alexhunter.dev

Australian Citizen; Willing to relocate or obtain Baseline / NV1 security clearance

SUMMARY

- Engineering graduate with First Class Honours and 2+ years of experience in internal system automation.
- Developed internal tools at Cochlear to increase efficiency by >50%.
- Built projects at UNSW which received full marks in autonomous robotics, computer vision, and software.

EDUCATION

Bachelor of Engineering in Mechatronic Engineering with Honours Class 1 / Bachelor of Science in Computer Science with Distinction – UNSW

2020 – 2025; Kensington, Sydney

- 87.89 WAM; Dean's Honours List (2020-2025); PhD scholarship offer
- High Distinctions including Advanced Autonomous Systems (98), Robot Design (96), Object Oriented Design (94), Software Construction (93), Robotics (92), Databases (90), Control Systems (90)

EXPERIENCE

Presales Engineer – Battery-X

2025 – 2026; Chatswood, Sydney

- Implemented a Shopify website, including available bundles and custom Liquid code to force users to log in before viewing the prices.
- Developed documentation and catalogues of all possible configurations, allowing sales teams and installers to easily communicate.

Student Engineer in Systems Verification & Validation – Cochlear Ltd.

2024 - 2025; Macquarie Park, Sydney

- Identified and reported essential product issues before release through formal verification processes, preventing public release of dangerous or unusable systems on several occasions.
- Helped redesign the importance reporting system to include safety, usability and commonality ratings, which prevented severely premature public releases while allowing releases with mild issues to meet deadlines.
- Built internal inventory management and test duration prediction tools which automated traceability reporting and test release estimation, reducing manual estimation effort 10× and total release verification duration by >50%.

- Ran upskilling workshops and 1-on-1 sessions, allowing team members to extend the automation toolchain independently.

Laboratory Demonstrator — UNSW

2021 – 2025; Kensington, Sydney

- Instructed Control Systems and Digital Circuit Design over 10 terms, facilitating lab sessions, assessing student performance, and received 90.9% positive student feedback.

PROJECTS

Wireless Wearable Phonocardiogram – UNSW Thesis

- Developed nRF52 SoC firmware to collect and transmit sound pressure data to an Android device, using custom BLE service and ADC.
- Created an Android app to receive and process data to display heart sounds, using Butterworth bandpass filter and custom processes.
- Rendered heart BPM and **distinct S1/S2 heart sounds** in real time

Autonomous Maze Solving Robot – UNSW Course (96%)

- Developed a fully closed loop control system to map the maze with **0% error** and traverse the maze in the **shortest path**. The robot was displayed at UNSW open day.
- Implemented a custom drive control, using cubic displacement trajectory and an adaptive yaw target to traverse multiple squares without stopping and had **0 crashes**.

Git Reimplemented in Python – UNSW Course (93%)

- Implemented executables for all major git commands (init, add, commit, log, show, rm, status, branch, checkout, merge), evaluated by course-designed autotests as 99.5% accurate.
- Developed a comprehensive shell test script suite, comparing real git commands to the corresponding reimplementation.

Extended Kalman Filter Vehicle Localiser – UNSW Course (98%)

Autonomous Robotic Arm (UR5e) – UNSW Course (92%)

TECHNICAL SKILLS

- **Software:** Git, Python, C/C++, React/React Nativ, JavaScript/TypeScript, HTML, CSS, SQL/PostgreSQL/PLpgSQL, Kotlin, Swift, REST APIs/HTTP, Object oriented design, JSON, XML, YAML, CI/CD, MATLAB, OpenCV
- **Hardware:** SolidWorks, Fusion360, LTSpice, Altium, Verilog/Xilinx
- **General:** Control systems and motion planning, debugging, Microsoft Teams/Word/Excel/PowerPoint, Agile/Scrum, documentation

REFERENCES

Available on request.